

LAHTI LAUNE, Finland

WMO: 029650

Lat: 60.9622N

Lon: 25.6308E

Elev: 79

StdP: 100.38

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.6	-21.1	-27.2	0.3	-24.4	-23.5	0.5	-21.0	5.2	-0.7	4.5	-1.1	0.4	10	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	27.5	18.7	25.6	17.7	23.8	16.9	20.2	25.0	19.0	23.7	17.9	22.3	2.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	22.5	17.3	12.5	21.1	16.2	11.6	19.9	58.5	25.1	54.4	23.7	50.8	22.2	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.5	4.0	3.5	DB	-27.3	29.4	4.4	2.1	-30.5	30.9	-33.1	32.2	-35.6	33.3	-38.8	34.8	(4)
(5)			WB	-27.5	21.3	4.3	1.4	-30.6	22.4	-33.1	23.2	-35.5	24.0	-38.7	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-6.5	-6.6	-2.7	3.8	9.6	14.0	17.1	15.4	10.3	4.5	0.1	-3.6	(6)	
	DBStd	9.40	6.63	6.76	4.71	3.65	4.05	3.33	3.03	2.96	3.29	4.05	4.54	6.05	(7)	
	HDD10.0	2548	510	465	393	189	56	4	0	2	37	173	297	421	(8)	
	HDD18.3	5017	768	698	651	437	270	135	61	99	242	428	547	679	(9)	
	CDD10.0	609	0	0	0	3	45	125	221	168	44	4	0	0	(10)	
	CDD18.3	37	0	0	0	0	1	6	23	8	0	0	0	0	(11)	
	CDH23.3	488	0	0	0	0	28	92	268	100	0	0	0	0	(12)	
	CDH26.7	87	0	0	0	0	3	12	59	13	0	0	0	0	(13)	
	Wind	WSAvg	1.5	1.6	1.5	1.6	1.6	1.5	1.4	1.3	1.3	1.5	1.6	1.7	(14)	
	Precipitation	PrecAvg	607	46	37	31	31	42	62	71	70	50	59	48	(15)	
(16)		PrecMax	807	86	82	51	59	94	101	179	143	131	149	101	(16)	
(17)		PrecMin	470	7	2	4	5	11	7	15	13	15	12	6	(17)	
(18)		PrecStd	82	18	20	14	15	21	26	39	31	29	31	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	5.2	10.8	20.0	26.4	28.3	30.6	28.4	22.0	14.9	9.7	7.8	(19)	
		MCWB	3.6	2.9	5.7	11.8	15.8	18.8	20.0	19.2	16.4	12.2	8.8	7.0	(20)	
	2%	DB	3.0	3.1	7.6	16.4	23.2	25.9	28.3	25.8	19.7	12.8	8.5	5.9	(21)	
		MCWB	2.5	2.2	3.7	9.2	14.5	17.1	19.6	18.1	15.0	11.2	7.9	5.2	(22)	
	5%	DB	2.1	2.1	5.1	13.8	20.8	23.9	26.4	24.1	17.8	11.7	7.3	4.6	(23)	
		MCWB	1.7	1.5	2.2	7.5	13.0	16.2	18.7	17.5	14.1	10.6	6.7	4.0	(24)	
	10%	DB	1.1	1.2	3.3	11.2	18.3	21.7	24.3	22.2	16.2	10.6	6.1	3.2	(25)	
		MCWB	0.7	0.8	1.4	6.3	11.6	15.0	17.7	16.7	13.5	9.7	5.6	2.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.4	6.4	12.3	18.0	20.4	22.0	21.1	16.8	12.9	9.1	7.0	(27)	
		MCDB	3.9	4.2	10.5	17.4	23.9	25.8	27.6	25.1	19.5	14.0	9.6	7.5	(28)	
	2%	WB	2.6	2.4	4.1	10.2	15.6	18.5	20.8	19.3	15.9	11.8	7.9	5.4	(29)	
		MCDB	2.9	2.9	6.7	15.3	20.5	23.6	26.2	23.7	18.4	12.5	8.5	5.8	(30)	
	5%	WB	1.7	1.6	2.8	8.4	13.9	17.1	19.8	18.3	14.9	10.8	6.8	4.1	(31)	
		MCDB	2.0	2.1	4.4	13.0	19.5	21.8	24.4	22.4	17.0	11.6	7.2	4.5	(32)	
	10%	WB	0.8	0.8	1.8	6.8	12.2	15.8	18.5	17.3	13.8	9.8	5.6	2.8	(33)	
		MCDB	1.0	1.2	3.0	10.5	17.3	20.5	23.0	21.0	15.8	10.5	6.0	3.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.5	8.7	10.3	11.6	10.6	10.6	10.7	9.3	6.4	4.3	4.9	(35)	
		MCDBR	4.0	4.8	11.2	15.5	16.4	15.0	14.3	14.5	11.9	7.6	4.6	4.2	(36)	
		MCWBR	4.0	4.4	7.9	9.0	8.2	7.1	6.5	7.6	7.7	5.8	4.3	4.1	(37)	
	5% WB	MCDBR	4.0	4.7	9.0	13.9	13.8	12.6	12.2	12.1	10.3	6.6	4.4	3.9	(38)	
		MCWBR	4.0	4.4	6.6	8.6	7.6	6.7	6.1	6.9	7.1	5.5	4.2	3.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.262	0.301	0.327	0.364	0.361	0.357	0.379	0.367	0.342	0.325	0.287	0.370	(40)	
	taud		2.099	2.164	2.236	2.341	2.410	2.448	2.402	2.430	2.493	2.400	2.191	2.602	(41)	
	Ebn at Noon		459	666	786	829	863	872	839	820	779	651	428	312	(42)	
	Edh at Noon		40	71	93	102	103	101	103	93	74	57	37	26	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.24	0.81	2.29	3.77	5.09	5.31	5.16	3.98	2.37	1.02	0.28	0.13	(44)	
	RadStd		0.04	0.18	0.29	0.39	0.56	0.41	0.50	0.44	0.31	0.17	0.05	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (59 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-148	-166	N/A	

Nomenclature: See separate page